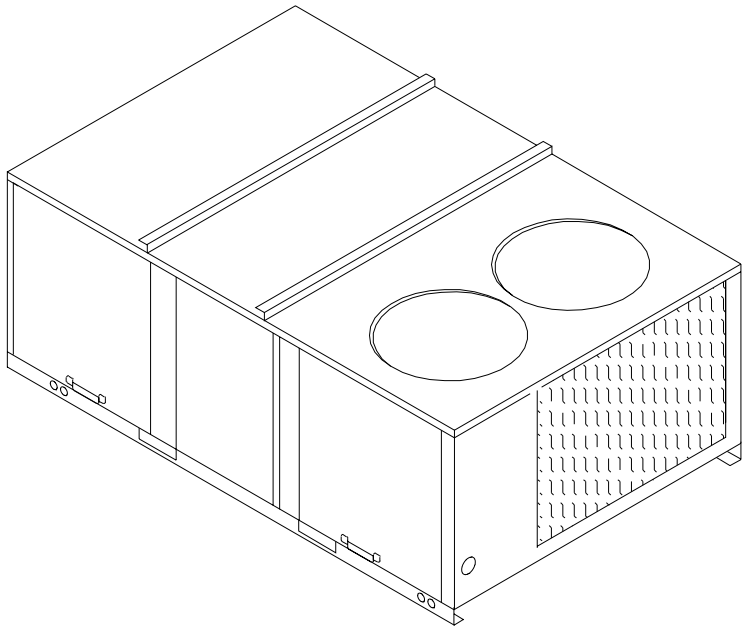
TC-SVF03B-EN

Service Facts

Packaged Cooling

15 Rooftop Units
with ReliaTel™ Controls



Models:

TCD180B300HB
TCD180B30AHB
TCD180B30BHB
TCD180B30CHB
TCD180B30FHB
TCD180B30GHB
TCH180B300HB
TCH180B30BHB
TCH180B30FHB
TCD180B400HB
TCD180B40AHB
TCD180B40BHB
TCD180B40CHB
TCD180B40FHB
TCD180B40GHB
TCH180B400HB
TCD180BW0FHB
TCD180BW0GHB
TCH180BW00HB
TCH180BW0BHB
TCH180BW0FHB

April 2005

Table of Contents

Sequence of Operation 3
Performance Data 5
Evaporator Fan Performance 6
Fan Speeds / Static Pressure Data 7
Pressure Curves 8
Refrigerant Circuit 9

NOTICE:
Warnings and Cautions appear at appropriate sections throughout this manual. Read these carefully.

WARNING– Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION – Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.

CAUTION – Indicates a situation that may result in equipment or property-damage-only accidents.

© 2003 American Standard Inc. All Rights Reserved

Sequence of Operation

⚠ WARNING
Live Electrical Components!

During installation, testing, servicing and troubleshooting of this product, it may be necessary to work with live electrical components. Have a qualified licensed electrician or other individual who has been properly trained in handling live electrical components perform these tasks. Failure to follow all electrical safety precautions when exposed to live electrical components could result in death or serious injury.

Sequence of Operation

These units are equipped with a microelectronic control feature, which provides operating functions that are significantly different than conventional units. It is referred to as the "ReliaTel Control System". The master module is the ReliaTel Refrigeration Module **(RTRM)**.

The RTRM provides compressor anti-short cycle timing functions through minimum "Off" and "On" timing to increase reliability, performance and to maximize unit efficiency.

Upon power initialization, the RTRM performs self-diagnostic checks to insure that all internal controls are functioning. It checks the configuration parameters against the components connected to the system.

The LED located on the RTRM module is turned "On" within one second after power-up if all internal

Cooling without an Economizer
When the system is in the "Cool" MODE, the RTRM energizes relay **(K3)**. When the normally open K3 contacts close, the switch over valve **(SOV)** is energized. When the zone temperature rises above the cooling setpoint controlband, the RTRM energizes the **(K9)** relay coil on the RTRM. When the K9 relay contacts close, the compressor contactor **(CC1)** coil is energized provided the low pressure control **(LPC1)** and **(HPC1)** are closed. When the CC1 contacts close, compressor **(CPR1)** and the outdoor fan motor **(ODM)** starts provided the normally open **ODF** relay contacts are closed, to maintain the zone temperature to within ± 2 F of the sensor setpoint at the sensed location.

Evaporator Fan Operation
When the fan selection switch is set to the "Auto" position, the RTRM energizes the **(K6)** relay coil approximately 1 second after energizing the compressor contactor coil **(CC1)** in the cooling mode. In the heating mode, the RTRM energizes the **(K6)** relay coil approximately 1 second before energizing the electric heat contactors. Closing the **K6** contacts on the RTRM energizes the indoor fan relay **(F)** coil to start the indoor fan motor **(IDM)**.

The RTRM de-energizes the fan relay **(F)** approximately 60 seconds after the cooling requirement has been satisfied to enhance unit efficiency. When the heating cycle is terminated, the indoor fan relay **(F)** coil is de-energized at the same time as the heater contactors.

When the fan selection switch is set to the "On" position, the RTRM keeps the indoor fan relay coil **(F)** energized for continuous fan motor operation.

When the unit is equipped with the optional fan failure switch, wired between terminals J7-10 and J7-11 on the ReliaTel Options Module **(RTOM)**, the **RTRM** produces an analog output if the fan failure switch **(FFS)** does not open within two minutes after a request for fan operation. When the system is connected to a remote panel, the "SERVICE" LED will be turned on when this failure occurs.

Low Ambient Operation
During low ambient operation, outside air temperature below 55 F, the RTRM will cycle the compressor and outdoor fan motor "Off" for approximately 3 minutes after every 10 minutes of accumulated compressor run time. The indoor fan motor **(IDM)** will continue to operate during this evaporator defrost cycle **(EDC)** and the compressor and outdoor fan will return to normal operation once the defrost cycle has terminated and the compressor "Off" time delay has been satisfied.

Cooling with an Economizer
When the system is in the "Cool" MODE, the RTRM energizes relay **(K3)**. When the normally open K3 contacts close, the switch over valve **(SOV)** is energized. The economizer is utilized to control the zone temperature providing the outside air conditions are suitable. Outside air is drawn into the unit through modulating dampers. When cooling is required and economizing is possible, the RTRM sends the cooling request to the unit economizer actuator **(ECA)** to open the economizer damper. The RTRM tries to cool the zone utilizing the economizer to slightly below the zone temperature setpoint. If the mixed air sensor **(MAS)** senses that the mixed air temperature is below 53 F, the damper modulates toward the closed position. If the zone temperature continues to rise above the zone temperature setpoint controlband and the economizer damper is fully open, the RTRM energizes the compressor contactor **(CC1)**.

The ECA continues to modulate the economizer damper open/closed to keep the mixed air temperature that is calculated by the RTRM.

If economizing is not possible, the ECA drives the damper to the minimum position setpoint when the indoor fan relay **(F)** is energized and allows mechanical cooling operation.

When the unit is equipped with the optional fan failure switch, wired between terminals J7-10 and J7-11 on the **RTOM**, the **RTRM** will stop all cooling functions and produce an analog output if the fan failure switch **(FFS)** does not open within 40 seconds after a request for fan operation. When the system is connected to a remote panel, the "SERVICE" LED will flash when this failure occurs.

Sequence of Operation

Economizer Set-Up
Adjusting the minimum position potentiometer located on the unit economizer actuator (ECA) sets the required amount of ventilation air.

Two of the three methods for determining the suitability of the outside air can be selected utilizing the enthalpy potentiometer on the ECA, as described below:

1. Ambient Temperature - controlling the economizing cycle by sensing the outside air-dry bulb temperature. The table below lists the selectable dry bulb values by potentiometer setting.

2. Reference Enthalpy - controlling the economizer cycle by sensing the outdoor air humidity. The table below lists the selectable enthalpy values by potentiometer setting. If the outside air enthalpy value is less than the selected value, the economizer is allowed to operate.

3. Comparative Enthalpy - By utilizing a humidity sensor and a temperature sensor in both the return air stream and the outdoor air stream, the unit control processor **(RTRM)** will be able to establish which conditions are best suited for maintaining the zone temperature, i.e. indoor conditions or outdoor conditions. The potentiometer located on the ECA is non-functional when both the temperature and humidity sensors are installed.

Potentiometer Setting	Dry Bulb	Enthalpy
A	73*	27
B	70	25
C	67	23
D	63	22

* Factory Setting

Heating Operation
When the system switch is set to the “Heat” MODE, the RTRM energizes relay **(K3)**. When the normally open K3 contacts open, the switch over valve **(SOV)** is de-energized. When the zone temperature falls below the heating setpoint controlband, the RTRM energizes the **(K9)** relay coil on the RTRM. When the K9 relay contacts close, the compressor contactor **(CC1)** coil is energized provided the low pressure control **(LPC1)** and **(HPC1)** are closed. When the CC1 contacts close, compressor **(CPR1)** and the outdoor fan motor **(ODM)** starts provided the normally open **ODF** relay contacts are closed, to maintain the zone temperature to within ± 2 F of the sensor setpoint at the sensed location.

The RTRM “Smart Recovery” function monitors the rate at which the zone temperature is changing every 9 minutes during the operating heating cycle. If the zone temperature is rising at a rate greater than 6 F per hour, no additional heat is requested (auxiliary electric heat). If not, the RTRM energizes the first stage auxiliary electric heat contactor **(AH)** (electric heat optional). A minimum of 10 seconds “Off” time must have elapsed between heater cycles. If mechanical heat and first stage auxiliary heat cannot provide the 6 F recovery rate, the RTRM energizes second stage auxiliary heat contactor **(BH)**, if applicable. The RTRM continues to monitor the rate of change and stages the electric heat “Off” as it determines that the mechanical heat (compressor operation) is sufficient.

Demand Defrost
Demand defrost is a standard feature which permits defrost whenever coil icing conditions begin to significantly reduce unit capacity. To permit defrost, the outdoor temperature must be below 52 F, coil temperature must be below 33 F, and the delta temperature F must exceed a RTRM calculated value. After 30 minutes of run time under defrost permit conditions, the RTRM initiates a defrost cycle. Upon termination of this cycle, the RTRM monitors the outdoor temperature **(ODT)** and the coil temperature **(CT)** and calculates the delta temperature F **(ODT - CT)**. This value is stored in memory and the RTRM calculates a defrost initiate value. The RTRM continually compares the delta temperature F to the defrost initiate value. Once the delta tee reaches the initiate value, a defrost cycle is initiated.

During the defrost cycle, the RTRM energizes the relay **(K3),** which energizes the switch over valve **(SOV)** through the normally open **K3** relay contacts. Then turns the outdoor fan motor **(ODM)** “Off” by energizing the **(K8)** relay, which energizes the **(ODF)** relay. The RTRM energizes the auxiliary electric heat contactor **(AH)** and **(BH)** (if applicable) if they are not operating, while maintaining compressor **(CPR1)** operation.

The defrost cycle is terminated based on the RTRM termination temperature calculation using the outdoor temperature **(ODT)** + 27 F. The defrost termination temperature **(DTT)** will typically be between 37 F and 52 F.

Emergency Heat Operation
When the system selection switch is in the “EM HEAT” MODE, and the zone temperature falls below the heating setpoint controlband, the RTRM bypasses compressor and outdoor fan operation and energizes the **K1** relay located on the RTRM. When **K1** relay contacts close, the first stage auxiliary electric heat contactor **(AH)** is energized. If the first stage of auxiliary electric heat can not satisfy the heating requirement, the RTRM energizes the **K2** relay coil located on the RTRM. When the **K2** relay contacts close, the second stage auxiliary electric heat contactor **(BH)** is energized. The RTRM cycles both the first and second stages of heat “On” and “Off” as required to maintain the zone temperature setpoint.

Performance Data

TC_180B	
A.R.I. Ratings	
Cooling Net Cap. BTUH (1)	174,000
Part Load	102,000
IPLV (4)	10.20
Indoor Air Flow (CFM)	5300
System Power (KW)	
Downflow/ Horizontal	17.57
EER (BTU per Watt) (2)	
Downflow/ Horizontal	9.9
OUTDOOR COIL — TYPE	Hi-Performance
Rows / F.P.I.	3/ 16
Face Area (Sq. Ft.)	27.12
Tube Size (In.)	0.375
INDOOR COIL–Type	Hi-Performance
Rows / F.P.I.	2/ 15
Face Area (Sq. Ft.)	17.5
Tube Size (In.)	0.375
Refrigerant Control Device	Short Orifice
Drain Conn. Size (In.)	1" NPT
REFRIGERANT	Charge (Lbs. of R-22) (3)
Circuit 1/ 2	17.5 / 9.5
WEIGHT	(Approx. Lbs)
Shipping/ Net	1866/ 1498
DIMENSIONS	H x W x L
Uncrated (In.)	50-2/16 x 70-13/16 x 106-11/16
FILTER	Furnished
Type	Throwaway
Filters Size (DF/ HZ)	20 X 25 X 2 (2) 20 x 25 x 2 (4)

VOLTS/ PHASE/ HZ	208-230/ 3/ 60	460/ 3/ 60	575/ 3/ 60
POWER CONNS			
Min. Circuit Ampacity			
Standard/ Oversize Motor	78 / 84	39/ 42	31/ 33
Fuse Size - Max. (Amps)			
Standard/ Oversize Motor	90 / 100	50/ 50	40/ 40
COMPRESSOR - No Used	Two	Two	Two
Volts/ Ph/ HZ	208-230/ 3/ 60	460/ 3/ 60	575/ 3/ 60
L.R. Amps (Circuit 1/ 2)	253 / 128	113/63	85/49
R.L. Amps (Circuit 1/ 2)	31.9 / 20.7	16.3/10	13.1/7.4
INDOOR FAN			
Type/ No. Used	FC Centrifugal/ 1	FC Centrifugal/ 1	FC Centrifugal/ 1
Dia. x Width (Inches)	15" x 15"	15" x 15"	15" x 15"
Drive/ No. Speeds	Belt/ Variable	Belt/ Variable	Belt/ Variable
INDOOR FAN MOTOR			
Motor HP/ RPM			
Standard Motor	3.0/ 1740	3.0/ 1740	3.0/ 1740
Oversize Motor	5.0/ 3450	5.0/ 3450	5.0/ 3450
Volts/ Ph/ Hz	208-230/ 3/ 60	460/ 3/ 60	575/ 3/ 60
L.R. Amps/ F.L. Amps			
Standard Motor	81/ 10.6	40.5/ 4.8	31/ 3.9
Oversize Motor	109.8/ 16.7	54.9/ 7.6	41.6/ 6.1
OUTDOOR FAN			
Type/ CFM	Propeller/ 10,200	Propeller/ 10,200	Propeller/ 10,200
No. Used/ Diameter	2/ 26	2/ 26	2/ 26
Drive/ No. Speeds	Direct/ 1	Direct/ 1	Direct/ 1
No. Motors/ HP/ RPM	2/ 0.50/ 1100	2/ 0.50/ 1100	2/ 0.50/ 1100
Volts/ Ph/ Hz	208-230/ 1/ 60	460/ 1/ 60	575/ 3/ 60
L.R. Amps/ F.L. Amps	8.8/ 3.2	3.8/ 1.6	3.2/ 1.3

Footnotes:

1. Cooling Performance is rated at 95 F ambient, 80 F entering dry bulb, 67 F entering wet bulb and nominal cfm listed.
   ARI capacity is net and includes the effect of fan motor heat.
   Base models are certified in accordance with the Unitary Large Equipment certification program,
   which is based on ARI Standard 340/360-93.
   All models tested in accordance with ARI Standard 340/360-93.
2. Rated at ARI conditions and in accordance with DOE test procedures.
3. Refrigerant charge shown is a nominal value; for a more precise value see the unit nameplate.
4. Intergrated Part Load Values is based on ARI Standard 340/360-93. Ratings are 80 F ambient, 80 F entering dry bulb,
   and 67 F entering wet bulb at ARI rated CFM.

Evaporator Fan Preformance

Evaporator Fan Performance — 15 Ton — TC*180B

	External Static Pressure (Inches of Water)																			
	.10		.20		.30		.40		.50		.60		.70		.80		.90		1.00	
CFM	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
			3 HP Standard Motor & Field Supplied Low Static Drive (1)								**3 HP Standard Motor & Sheaves**									
4800	—	—	563	1.17	596	1.29	626	1.41	656	1.52	685	1.63	714	1.76	742	1.90	768	2.04	794	2.18
5400	585	1.47	618	1.60	648	1.72	676	1.86	703	1.99	729	2.12	756	2.25	782	2.38	807	2.52	832	2.68
6000	643	1.99	673	2.12	701	2.26	727	2.41	752	2.56	777	2.71	801	2.85	825	2.99	848	3.13	871	3.27
6600	701	2.61	729	2.76	755	2.91	780	3.07	804	3.23	827	3.39	849	3.56	871	3.72	892	3.88	914	4.03
7200	760	3.35	786	3.52	810	3.69	834	3.85	856	4.02	878	4.20	899	4.38	919	4.56	939	4.74	959	4.91
			(2)		**5 HP Oversized Motor & Sheaves**															

Evaporator Fan Performance — 15 Ton — TC*180B — Continued

	External Static Pressure (Inches of Water)																			
	1.10		1.20		1.30		1.40		1.50		1.60		1.70		1.80		1.90		2.00	
CFM	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
	3 HP Standard Motor & Sheaves						**3 HP Standard Motor & High Static Drive Accessory**													
4800	819	2.30	844	2.43	871	2.57	898	2.70	923	2.83	946	2.96	970	3.09	994	3.23	1018	3.39	1041	3.54
5400	855	2.84	878	3.00	901	3.15	923	3.29	945	3.43	969	3.58	993	3.73	1016	3.88	1038	4.03	1060	4.17
6000	894	3.43	916	3.61	938	3.79	959	3.97	980	4.15	1000	4.31	1019	4.46	1040	4.62	1061	4.78	1082	4.95
6600	935	4.18	956	4.34	977	4.51	997	4.70	1018	4.90	1037	5.10	1056	5.30	1075	5.49	1093	5.67	—	—
7200	979	5.08	999	5.25	1018	5.41	1038	5.58	1057	5.77	—	—	—	—	—	—	—	—	—	—
	5 HP Oversized Motor & Sheaves										**5 HP OS Motor & Field Supplied Drive (3)**									

For Standard Evaporator Fan Speed (RPM), reference Table PD-22.
For Standard Motor & High Static Drive Accessory Fan Speed (RPM), reference Table PD-23.
For Oversized Evaporator Fan Speed (RPM), reference Table PD-24.

NOTES:
Fan motor heat (MBH) = 3.15 x Fan BHP
Data includes pressure drop due to wet coils and filters
No accessories or options are included in pressure drop data.
Refer to Table PD-26 to determine add'l static pressure drop due to other options/accessories.

1. Field Supplied Fan Sheave BK105 Required. Field Supplied Belt may be necessary.
2. 5 HP Oversized Motor with Field Supplied Fan Sheave BK160 Required.
   Field Supplied Belt may be necessary.
3. 5 HP Oversized Motor with Field Supplied Motor Sheave IVP50 Required. Field Supplied Belt may be necessary.
4. For 380V/60 Hz units, the oversized motor is used as the standard motor. Refer to "Motor & Sheave/Fan Speed" tables to determine operating range.

Trane's factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Fan Speeds / Static Pressure Data

Standard Motor And Sheave/Fan Speed (RPM)

Unit Model No.	6 Turns Open	5 Turns Open	4 Turns Open	3 Turns Open	2 Turns Open	1 Turns Open	Closed
*180	679	721	764	806	849	891	N/A

Standard Motor And High Static Drive Accessory/Fan Speed (RPM)

Unit Model No.	6 Turns Open	5 Turns Open	4 Turns Open	3 Turns Open	2 Turns Open	1 Turns Open	Closed
*180	806	849	891	934	976	1019	N/A

Oversized Motor And Sheave/Fan Speed (RPM)

Unit Model No.	6 Turns Open	5 Turns Open	4 Turns Open	3 Turns Open	2 Turns Open	1 Turns Open	Closed
*180	807	856	907	957	1008	1068	N/A

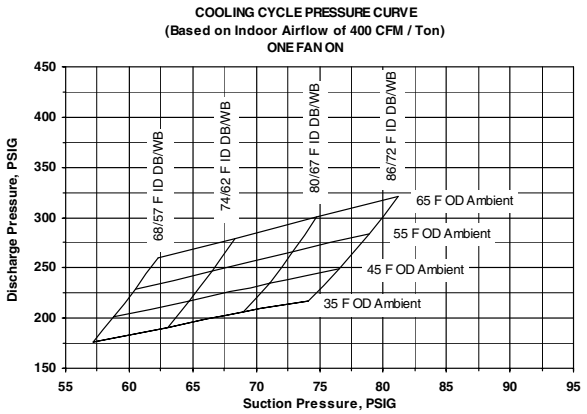
Static Pressure Drops Through Accessories — $H_2 0$

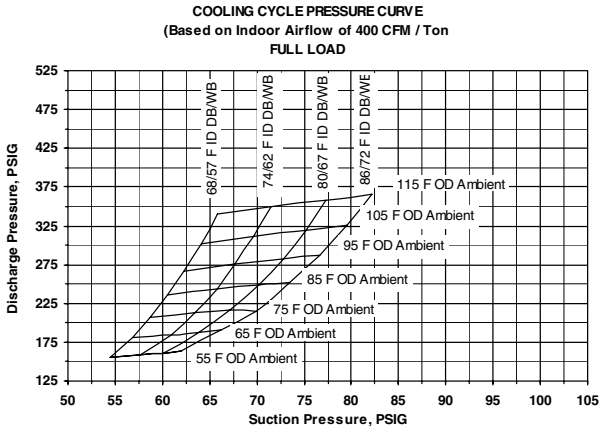
Unit Model	Airflow No.m^3/h	Standard Filters	Pleated Filters	Economizer with OA/RA Dampers (1) 100% OA	100% RA	Electric Heater - Nom kW (2) 14-23	36	54	72
*180	4800	.07	.10	.25	.03	.04	.04	.05	—
*180	6000	.10	.15	.33	.04	.06	.07	.08	—
*180	7200	.15	.19	.40	.05	.09	.10	.12	—

Notes:
1. OA = Outside Air and RA = Return Air
2. Nominal kW ratings are at 460V.
* Indicates both downflow and horizontal units

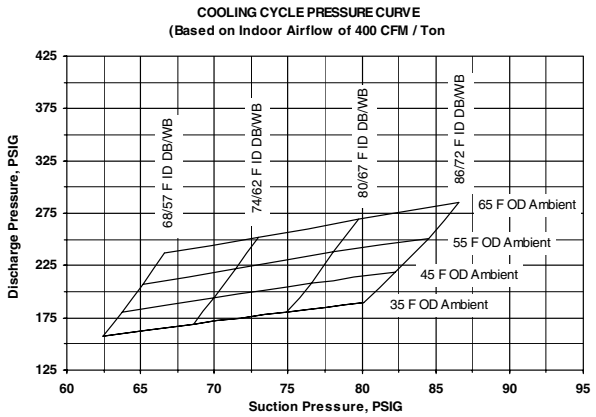
Pressure Curves

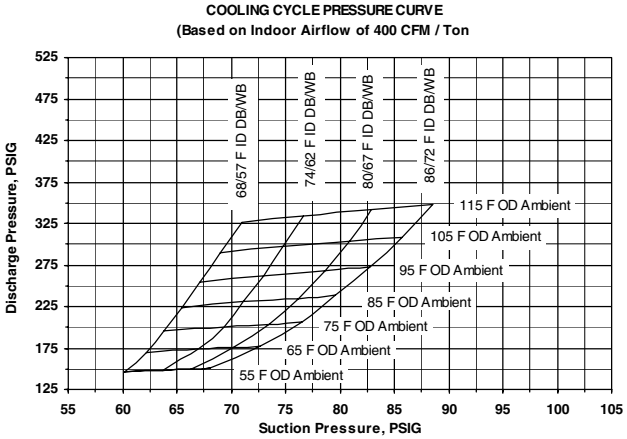
Circuit # 1

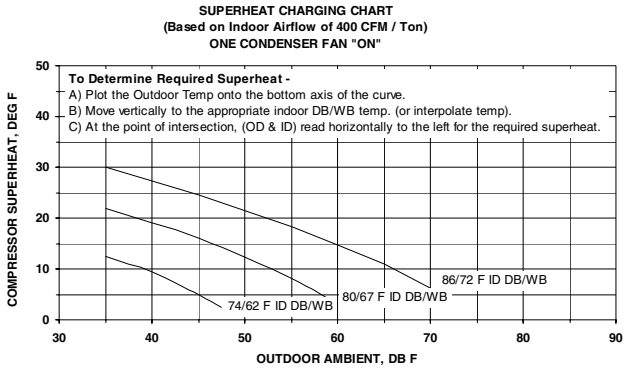


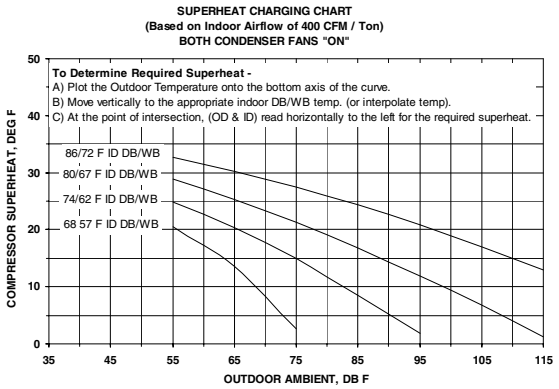


Circuit # 2







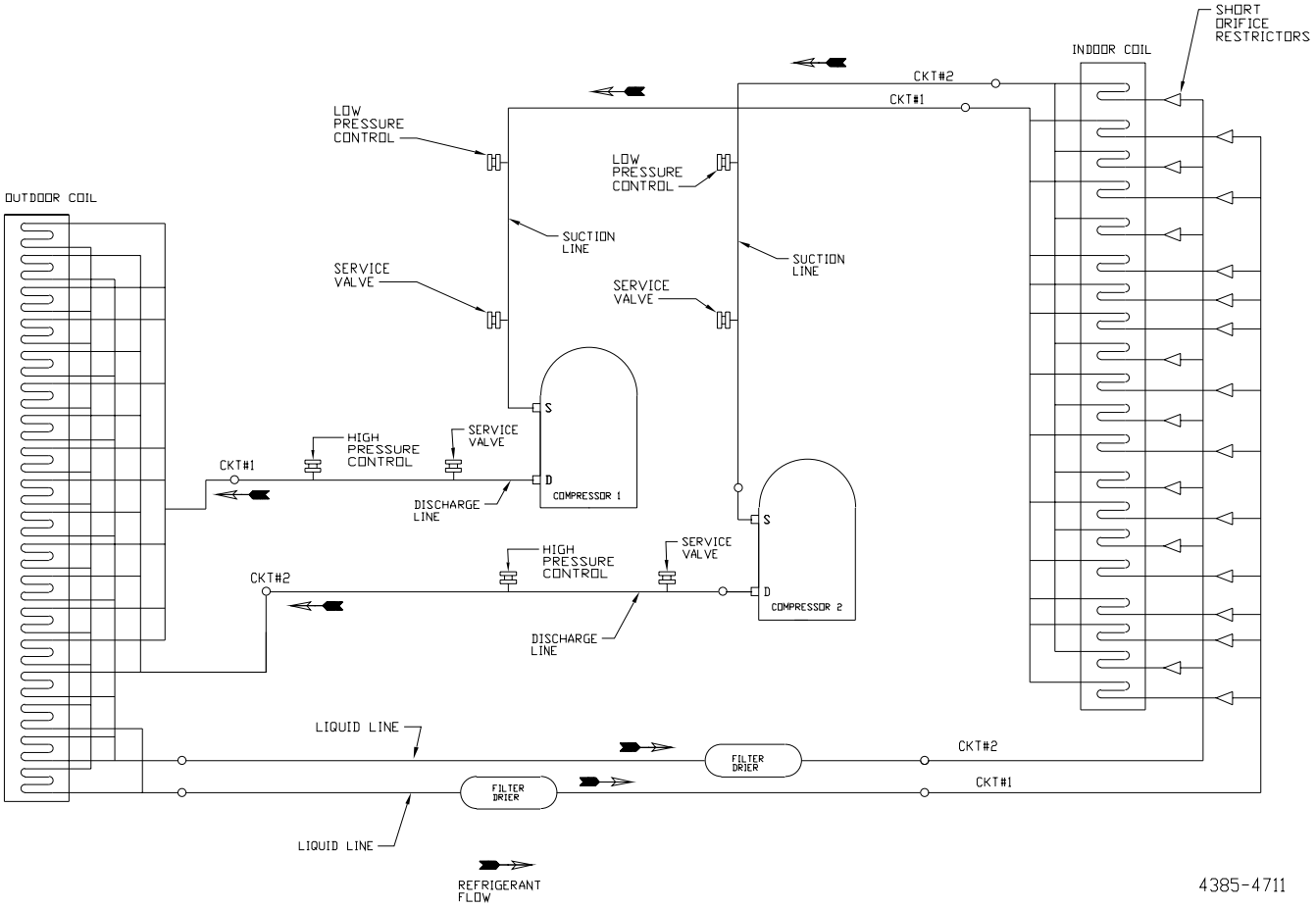


1) REFRIGERANT CHARGE - ADD if the superheat is more than 5 F above curve value.
- REDUCE if the superheat is more than 5 F below curve value.
- OK if the superheat is within 5 F of curve value.
2) Do not add refrigerant if the superheat is less than 5 F.
3) Curves are based on 400 CFM/Ton Indoor Airflow @ 50% R.H.
4) System must be running at stablized conditions before measuring superheat.

To Check Operating Pressures:

1. Start the unit and allow the pressures to stabilize.
2. Measure the indoor DB/WB temperature entering the indoor coil.
3. Measure the outdoor air dry bulb temperature.
4. Take discharge and suction pressure readings.
5. Plot the outdoor dry bulb and the indoor DB/WD temperature onto the chart.
6. At the point of intersection, read down for the suction pressure and to the left for the discharge pressure.

Refrigerant Circuit



Literature Order Number	TC-SVF03B-EN
File Number	SV-UN-TC-SVF03B-EN 04/05
Supersedes	New
Stocking Location	Webb Mason

The manufacturer has a policy of continuous product and product data improvement and reserves the right to change design and specifications without notice.